Prediction Model In Class

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PUBLISHED
January 1, 2025

I. Executive Summary

Define Focus of Prediction

What are you trying to predict?
Ans: We are trying to predict house sales price.

What type of prediction model is it?
Ans: It is linear regression prediction model.

Communicate Findings

What were your findings from your prediction model?

Findings:

- Overall the model performs well.
- Predictions are quite close to actual price.
- Accuracy drops when it comes to higher price range.
- Most points follow the diagonal line.
- Neighborhood, overall quality, and Ground living area are the most influential predictors.

How do these findings tie back to question or problem at hand?

Findings:

- This model effectively predicts house prices.
- The model shows a comprehensible data trend even though the accuracy drops at higher price range.

Take Action

on the analysis, what do actions do you recommend

Actions:(Good models)

- Add more variables with higher prices to get more accurate result on higher price range.
- Increase the sample size to increase the overall accuracy of the model.
- Try models with different predictors.

Actions:(Bad model, lm_t5)

- Use more relevant predictors to make model.
- Avoid too simple variables.

II. Prediction Model

Summarization of the model

- The outcome variable is sale price.

-The models consists of Neighborhood, Overall quality, Ground living area, Garage cars, lot area.

- The good models includes Neighborhood, Overall quality, and Ground living area.
- The bad model has Garage cars and lot area.

Overall, the models are accurate and can predict house prices with the exception of high price range houses.

Collect Data

We are using the Ames Housing data.

Clear Environment

Load Libraries and Custom Functions

Load Data

Clean Data

Convert Data to Useful Definitions

Modify Data - Create Calculated Fields

We will build two different models with the Ames housing data set. The first model will use regression to predict the house price - a continuous value. The 2nd

MAKE THE MODEL TO GET THE SIGN FACTORS

CHECK THE WHOLE DATA

Model 1: Linear Regression Data

Handle Missing Values

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```

Normalize Predictor Variables

This is done so that large numeric values (like lot area) doesn't overshadow the impact of small number (like Full Bath). This puts all numerical values on a standard scale.

Create the Training & Test Data Sets

Training data sets are to develop a base model. This is typically 60-80% of the data set. The reason is to have large enough data set to build a quality model but without "overfitting".

Test data set is used to prove out the quality of the model. This is typically 40-20% of the original data set.

Overfitting is when you over engineer your model to the data set only and not for general purpose usage across data sets.

###Check for the Significant factors

Test some Models to find the significant ones

Build and Evaluate 3 Distinct Models

```
Call:
```

```
lm(formula = Sale_Price ~ Neighborhood + Overall_Qual, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-239930  -19801   -2419   15856  267383 

Coefficients:
                                                      Estimate Std. Error t value
(Intercept)                                              56122      22489   2.496
NeighborhoodCollege_Creek                                17025       3780   4.503
NeighborhoodOld_Town                                    -23110       3716  -6.219
NeighborhoodEdwards                                     -11618       4002  -2.903
NeighborhoodSomerset                                     17114       4675   3.661
NeighborhoodNorthridge_Heights                           49486       5344   9.260
NeighborhoodGilbert                                      12829       4525   2.835
NeighborhoodSawyer                                       -2795       4505  -0.620
NeighborhoodNorthwest_Ames                               16528       4995   3.309
NeighborhoodSawyer_West                                  11917       4818   2.474
NeighborhoodMitchell                                      9302       4799   1.938
NeighborhoodBrookside                                   -16099       4819  -3.341
NeighborhoodCrawford                                     35163       5303   6.631
NeighborhoodIowa_DOT_and_Rail_Road                      -27248       5207  -5.233
NeighborhoodTimberland                                   37026       6149   6.022
NeighborhoodNorthridge                                   91076       6661  13.673
NeighborhoodStone_Brook                                  51935       7310   7.105
NeighborhoodSouth_and_West_of_Iowa_State_University     -16419       7380  -2.225
NeighborhoodClear_Creek                                  47686       7518   6.343
NeighborhoodMeadow_Village                              -29688       8144  -3.645
NeighborhoodBriardale                                   -49991       8947  -5.587
NeighborhoodBloomington_Heights                           4388       8837   0.497
NeighborhoodVeenker                                      31986       9534   3.355
NeighborhoodNorthpark_Villa                             -25122       9981  -2.517
NeighborhoodBlueste                                     -40662      13913  -2.923
NeighborhoodGreens                                      -38715      16299  -2.375
NeighborhoodGreen_Hills                                 143103      38670   3.701
NeighborhoodLandmark                                    -22235      38621  -0.576
Overall_QualPoor                                         20650      25752   0.802
Overall_QualFair                                         41440      23378   1.773
Overall_QualBelow_Average                                61695      22520   2.740
Overall_QualAverage                                      81298      22415   3.627
Overall_QualAbove_Average                               103113      22438   4.596
Overall_QualGood                                        130776      22520   5.807
Overall_QualVery_Good                                   179552      22681   7.917
Overall_QualExcellent                                   272428      23146  11.770
Overall_QualVery_Excellent                              355427      23853  14.901
                                                      Pr(>|t|)
(Intercept)                                           0.012656 *
NeighborhoodCollege_Creek                             7.07e-06 ***
NeighborhoodOld_Town                                  6.06e-10 ***
NeighborhoodEdwards                                   0.003732 **
NeighborhoodSomerset                                  0.000258 ***
```

```
NeighborhoodNorthridge_Heights                       < 2e-16 ***
NeighborhoodGilbert                                 0.004625 **
NeighborhoodSawyer                                  0.535044
NeighborhoodNorthwest_Ames                          0.000953 ***
NeighborhoodSawyer_West                             0.013455 *
NeighborhoodMitchell                                0.052722 .
NeighborhoodBrookside                               0.000851 ***
NeighborhoodCrawford                                4.27e-11 ***
NeighborhoodIowa_DOT_and_Rail_Road                  1.84e-07 ***
NeighborhoodTimberland                              2.05e-09 ***
NeighborhoodNorthridge                               < 2e-16 ***
NeighborhoodStone_Brook                             1.66e-12 ***
NeighborhoodSouth_and_West_of_Iowa_State_University 0.026201 *
NeighborhoodClear_Creek                             2.77e-10 ***
NeighborhoodMeadow_Village                          0.000274 ***
NeighborhoodBriardale                               2.62e-08 ***
NeighborhoodBloomington_Heights                     0.619530
NeighborhoodVeenker                                 0.000808 ***
NeighborhoodNorthpark_Villa                         0.011914 *
NeighborhoodBlueste                                 0.003510 **
NeighborhoodGreens                                  0.017630 *
NeighborhoodGreen_Hills                             0.000221 ***
NeighborhoodLandmark                                0.564871
Overall_QualPoor                                    0.422707
Overall_QualFair                                    0.076445 .
Overall_QualBelow_Average                           0.006206 **
Overall_QualAverage                                 0.000294 ***
Overall_QualAbove_Average                           4.58e-06 ***
Overall_QualGood                                    7.37e-09 ***
Overall_QualVery_Good                               4.00e-15 ***
Overall_QualExcellent                                < 2e-16 ***
Overall_QualVery_Excellent                           < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 38530 on 2014 degrees of freedom
Multiple R-squared:  0.7735,    Adjusted R-squared:  0.7694
F-statistic:   191 on 36 and 2014 DF,  p-value: < 2.2e-16


Call:
lm(formula = Sale_Price ~ Overall_Qual + Gr_Liv_Area, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-446068  -19806     679   18076  223149

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                   82578      21698   3.806 0.000146 ***
Overall_QualPoor              12696      24980   0.508 0.611345
```

```
Overall_QualFair              20463      22702   0.901 0.367490
Overall_QualBelow_Average     40564      21848   1.857 0.063504 .
Overall_QualAverage           64985      21722   2.992 0.002808 **
Overall_QualAbove_Average     81705      21754   3.756 0.000178 ***
Overall_QualGood             113127      21802   5.189 2.33e-07 ***
Overall_QualVery_Good        167512      21908   7.646 3.17e-14 ***
Overall_QualExcellent        260003      22308  11.655  < 2e-16 ***
Overall_QualVery_Excellent   303378      23231  13.059  < 2e-16 ***
Gr_Liv_Area                   26862       1012  26.542  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37470 on 2040 degrees of freedom
Multiple R-squared:  0.783, Adjusted R-squared:  0.7819
F-statistic:   736 on 10 and 2040 DF,  p-value: < 2.2e-16


Call:
lm(formula = Sale_Price ~ Lot_Area + Garage_Cars, data = train_data)

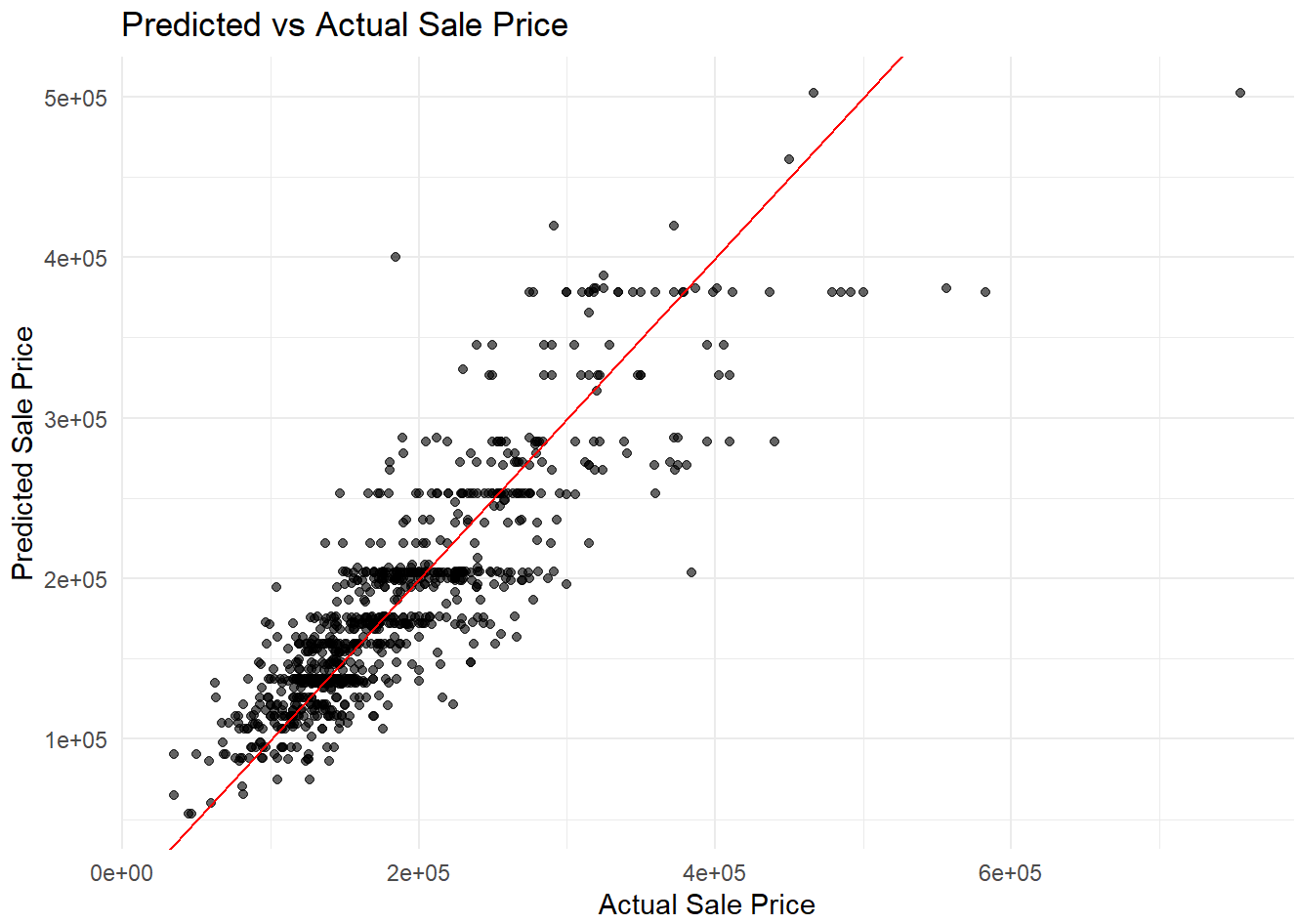
Residuals:
    Min      1Q  Median      3Q     Max
-260301  -35142   -6340   23979  477586

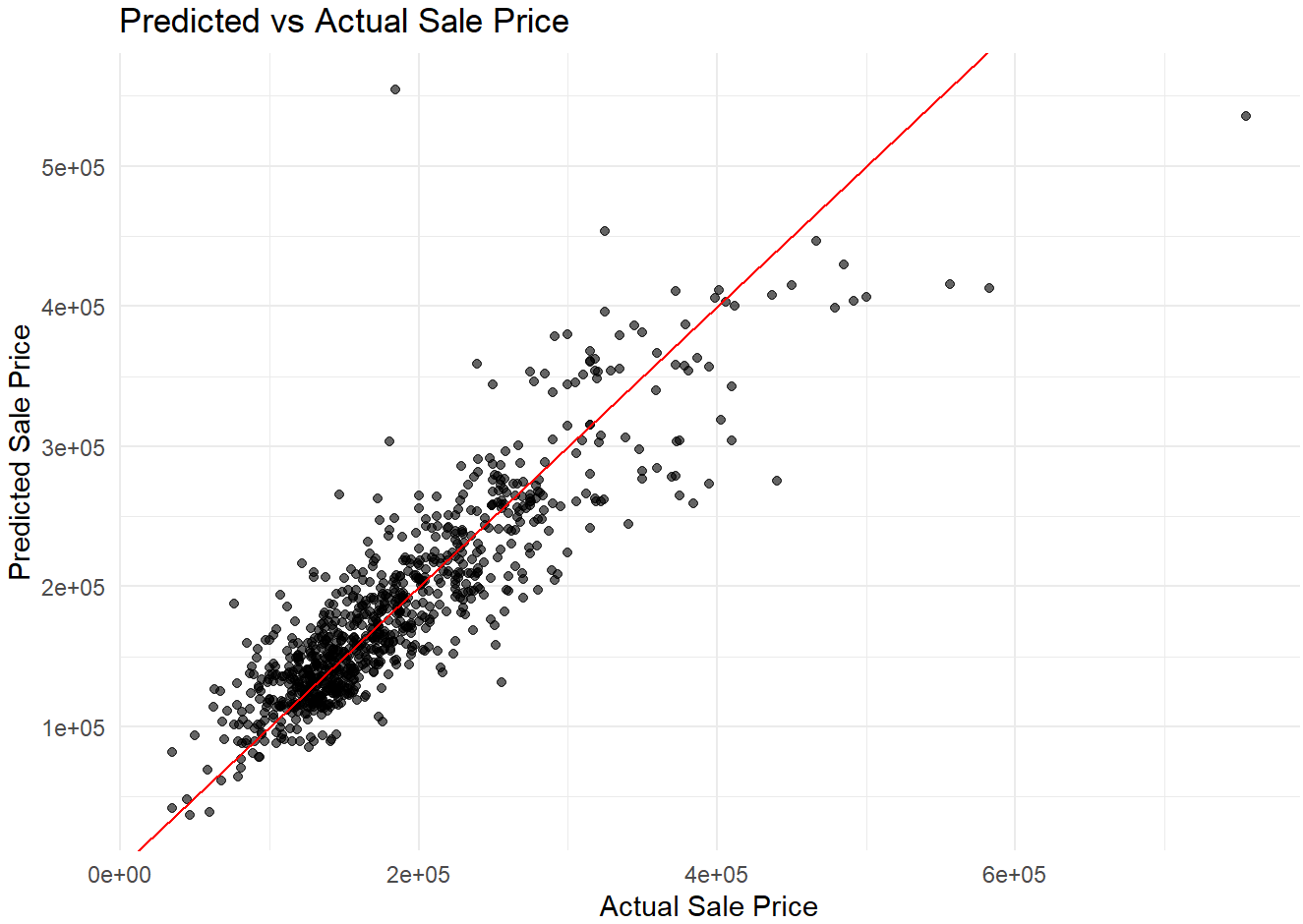
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   179507       1334  134.54   <2e-16 ***
Lot_Area       11482       1240    9.26   <2e-16 ***
Garage_Cars    49310       1358   36.32   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

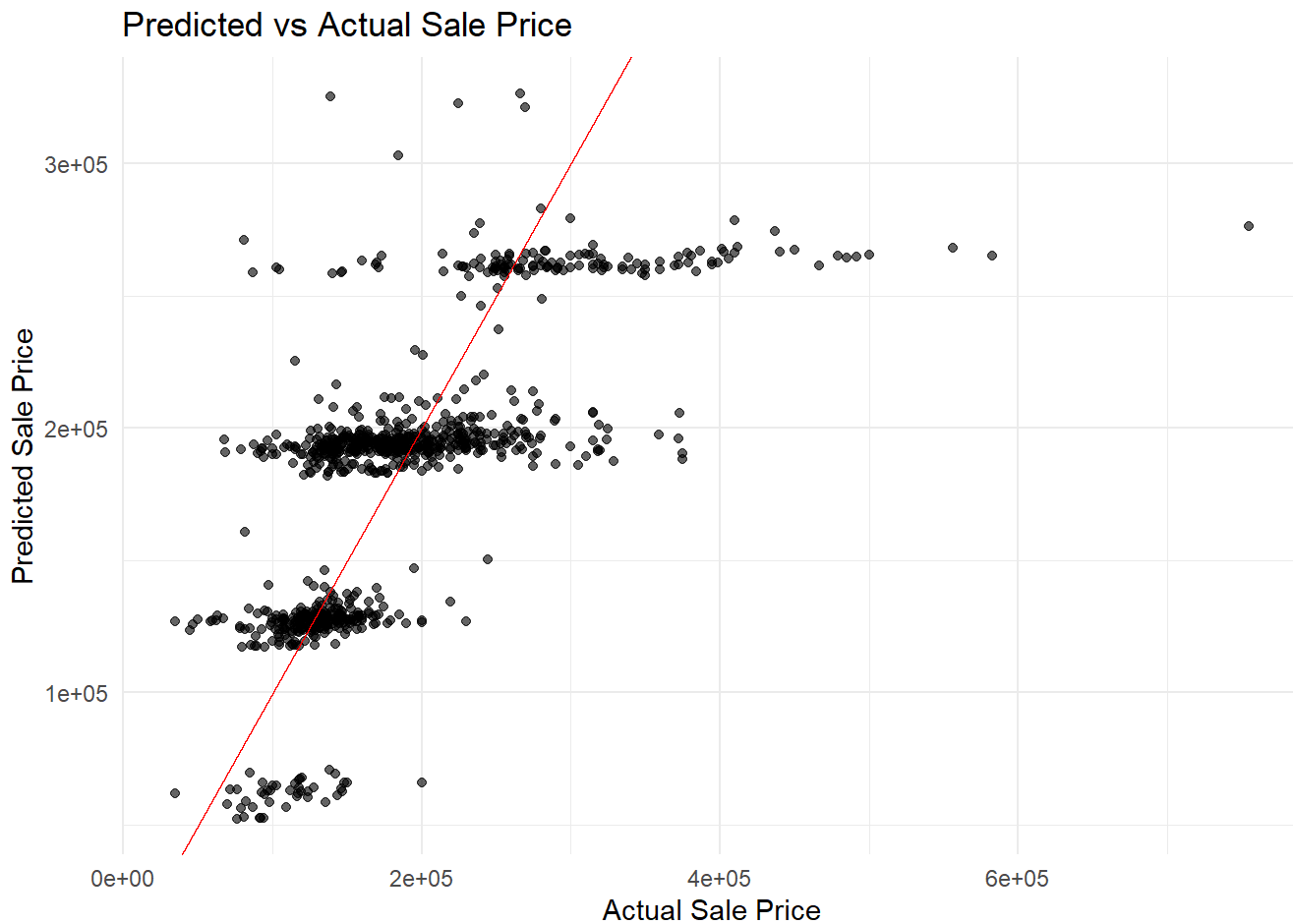
Residual standard error: 60420 on 2048 degrees of freedom
Multiple R-squared:  0.4334,    Adjusted R-squared:  0.4329
F-statistic: 783.4 on 2 and 2048 DF,  p-value: < 2.2e-16
```

Make a prediction and evaluate model quality

The prediction uses the fitted model (meaning a model derived from the training data set) and makes a prediction from the Test data set.


Predicted vs Actual Sale Price
Predicted Sale Price
5e+05
4e+05
3e+05
2e+05
1e+05
0e+00
2e+05
4e+05
6e+05
Actual Sale Price


Predicted vs Actual Sale Price
Predicted Sale Price
5e+05
4e+05
3e+05
2e+05
1e+05
0e+00
2e+05
4e+05
6e+05
Actual Sale Price



Evaluate the Model

##lm_t1 = Neighborhood and Overall quality ##lm_t2 = Overall quality and Ground living area ##lm_t3 = Lot area and Garage car

```
# Indices of model performance

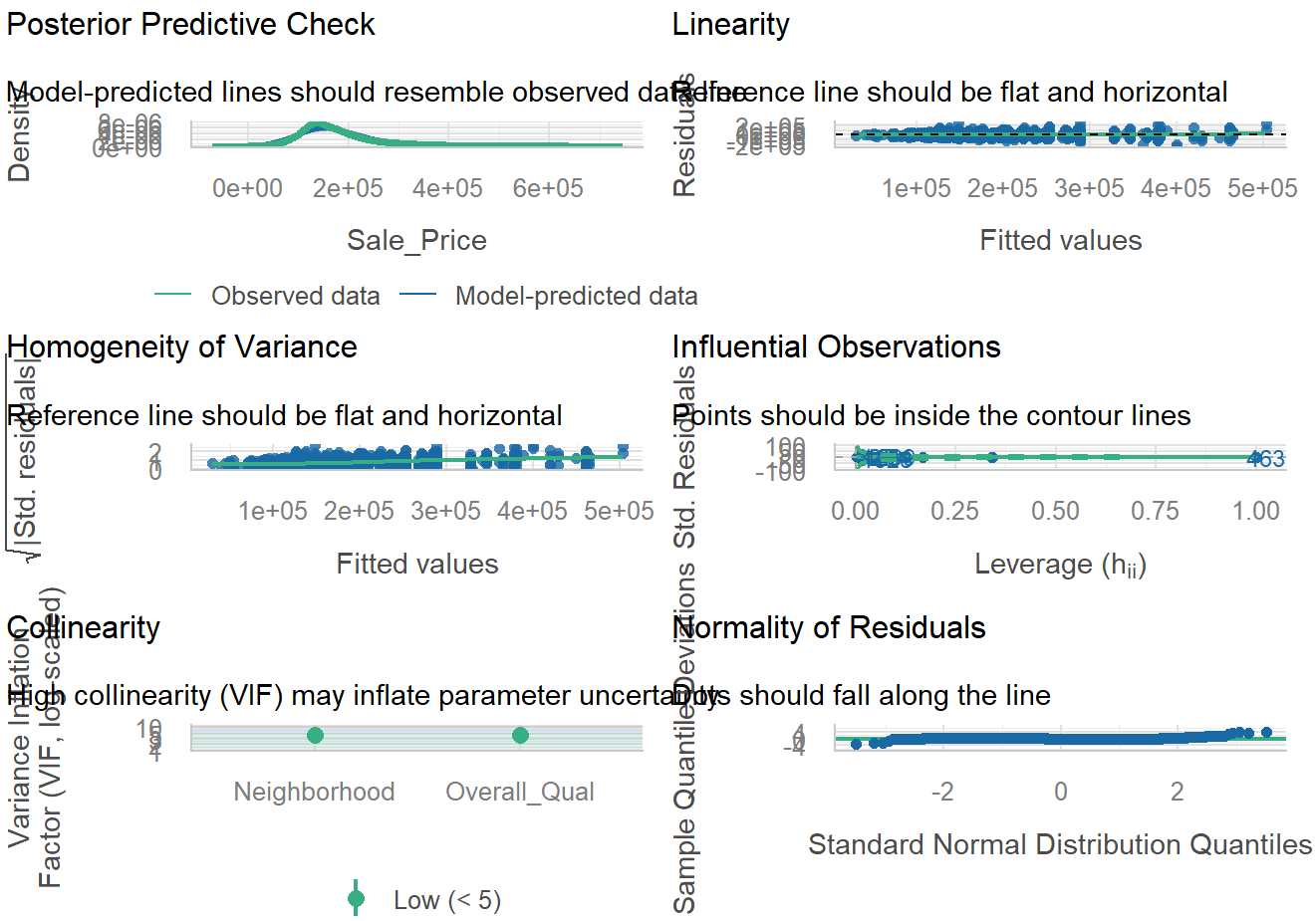
AIC     |    AICc |     BIC |    R2 | R2 (adj.) |      RMSE |     Sigma
-----------------------------------------------------------------------
49172.5 | 49173.9 | 49386.3 | 0.773 |     0.769 | 38176.290 | 38525.370

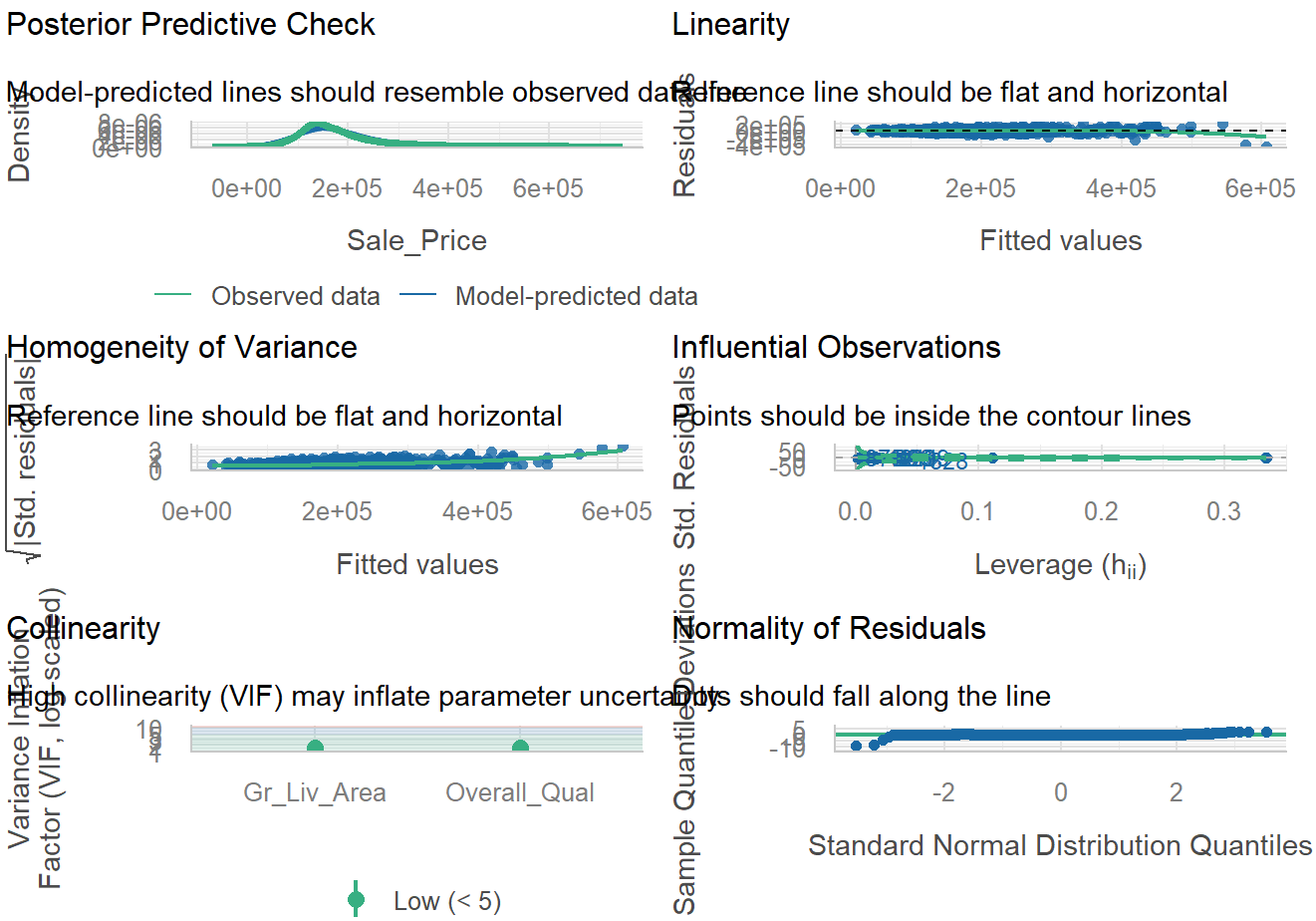
# Indices of model performance

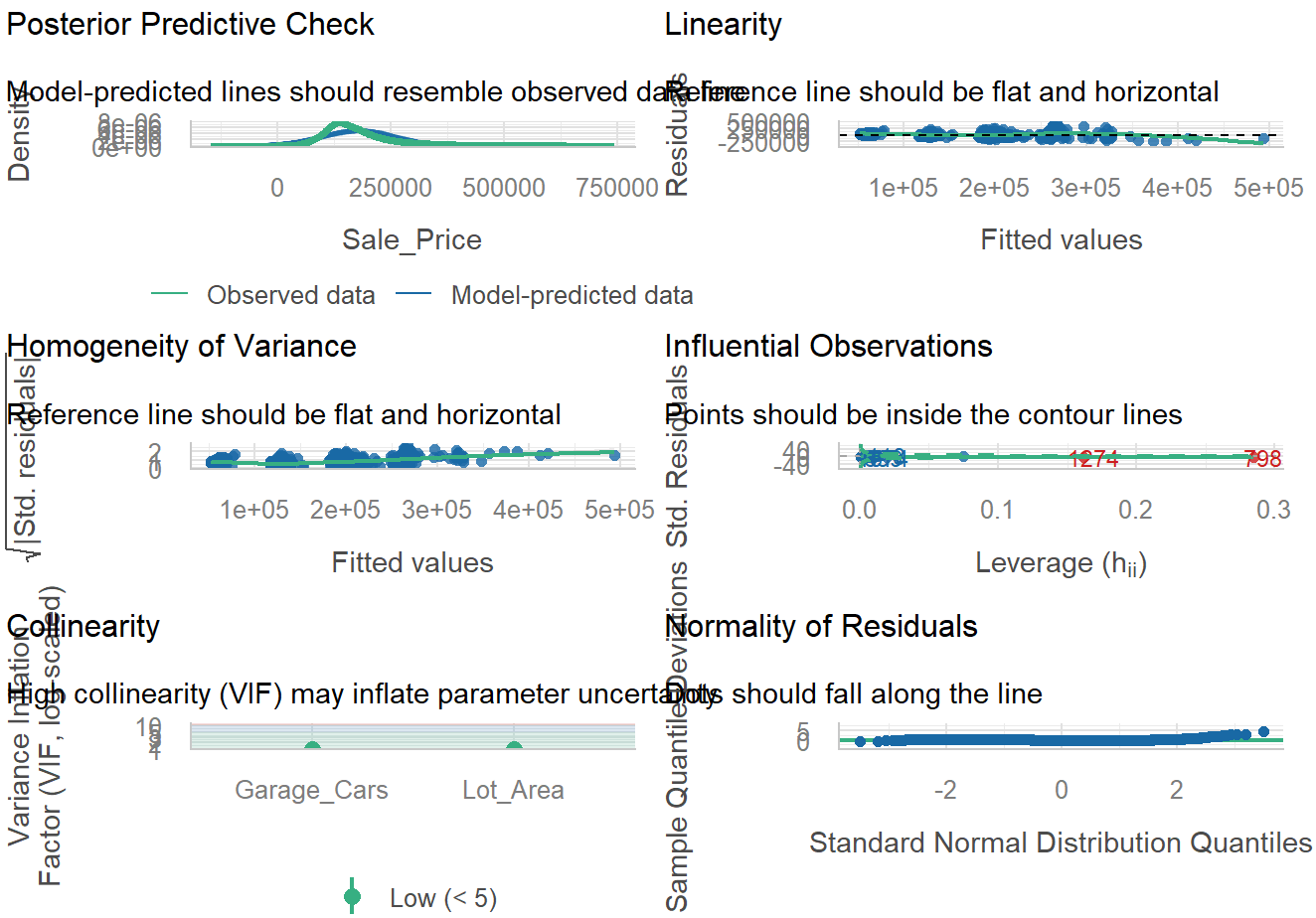
AIC     |    AICc |     BIC |    R2 | R2 (adj.) |      RMSE |     Sigma
-----------------------------------------------------------------------
49032.7 | 49032.9 | 49100.2 | 0.783 |     0.782 | 37368.383 | 37468.995

# Indices of model performance

AIC     |    AICc |     BIC |    R2 | R2 (adj.) |      RMSE |     Sigma
-----------------------------------------------------------------------
50984.8 | 50984.8 | 51007.3 | 0.433 |     0.433 | 60377.416 | 60421.621
```


Posterior Predictive Check
Model-predicted lines should resemble observed data
Density
0e+00
2e+05
4e+05
6e+05
Sale_Price
Observed data
Model-predicted data
Linearity
Reference line should be flat and horizontal
Residuals
1e+05
2e+05
3e+05
4e+05
5e+05
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
463
0.00
0.25
0.50
0.75
1.00
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Neighborhood
Overall_Qual
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles Deviations
-2
0
2
Standard Normal Distribution Quantiles


Posterior Predictive Check
Model-predicted lines should resemble observed data
Density
0e+00
2e+05
4e+05
6e+05
Sale_Price
Observed data
Model-predicted data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
0.0
0.1
0.2
0.3
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Gr_Liv_Area
Overall_Qual
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
-2
0
2
Standard Normal Distribution Quantiles



Here is the checklist to evaluate a model from the “check_model()” method of the Performance library.

##Findings(Neighborhood and Overall quality):

- **Posterior Predictive Check** - The prediction values align with the observed values for the most part but on the top there are some extreme values.
- **Linearity** - Residuals are scatter randomly. Line is flat and horizontal with a bit of exception at higher value. There is no systematic pattern.
- **Homogeneity of Variance (Homoscedacity)** - The spread of residuals is not consistent across fitted values. As is goes more extreme the residuals are increased slightly on higher fitted values.
- **Influential Observations** - Most of the points are in the contour execpt one point at 463.
- **Collinearity** - There is no strong collinearity. VIF is less than 5 which is best.
- **Normality of Residuals** - The residuals are not normally distributed. Points are deviated from reference line.

##Findings(Overall quality and Ground living area):

- **Posterior Predictive Check** - The prediction values align with the observed values for the most part but on the top there are some extreme values.

- **Linearity** - Residuals are scatter randomly. Line is flat and horizontal for most part and is slightly downward at high value.
- **Homogeneity of Variance (Homoscedacity)** - The spread of residuals are consistent across fitted values and consistently increases as in approaches to higher value.
- **Influential Observations** - Most of the points are in the contour expect some extreme values.
- **Collinearity** - VIF 1.5 for both indicating low collinearity and no significant collinearity among predictors.
- **Normality of Residuals** - The residuals are not normally distributed. Points are deviated from reference line in the beginning and at the end.

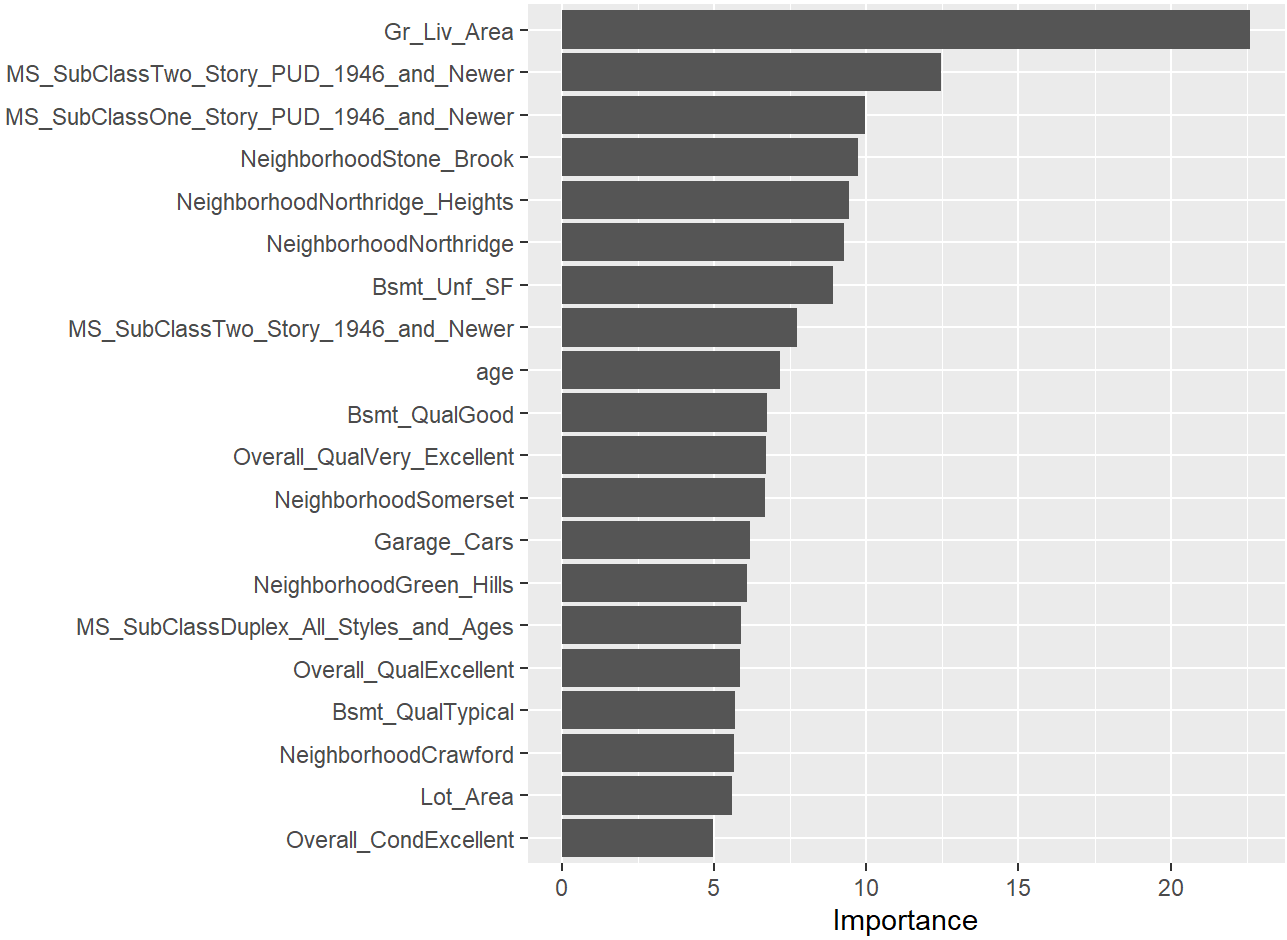
##Findings(Lot area and Garage car):

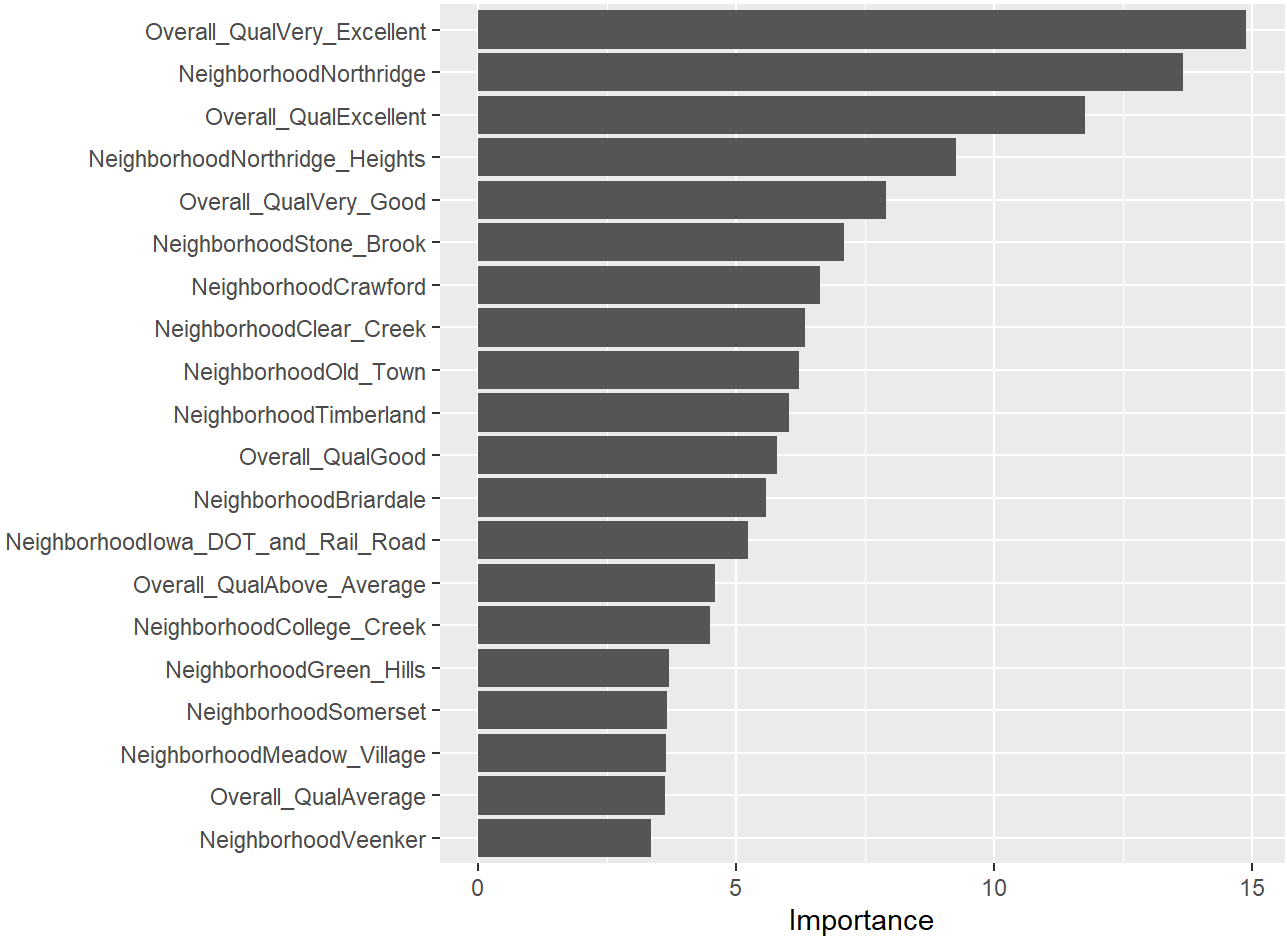
- **Posterior Predictive Check** - The prediction values doesn't align with the observed values at around 0-300k.
- **Linearity** - Residuals are scatter randomly. Line doesn't align with the horizontal line.
- **Homogeneity of Variance (Homoscedacity)** - The spread of residuals is not consistent across fitted values. As is goes more extreme the residuals line decreases and increases like funnel shape.
- **Influential Observations** - There are multiple points that influences the contour and the most extreme is 798.
- **Collinearity** - VIF is 1 for both indicating there are no strong collinearity among the predictors.
- **Normality of Residuals** - The residuals are not normally distributed. Points are deviated from reference line at both tails.

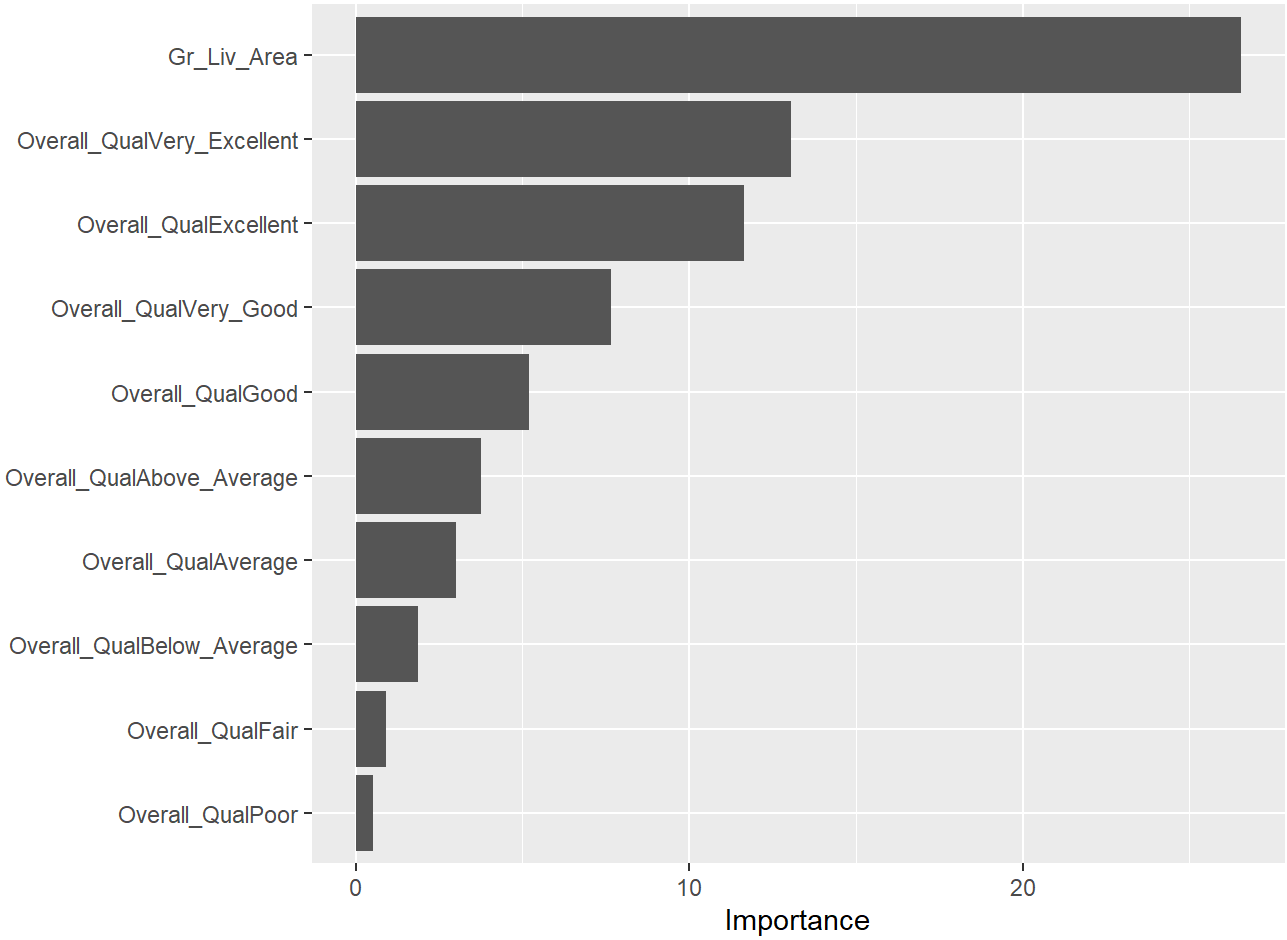
Check Variable Importance

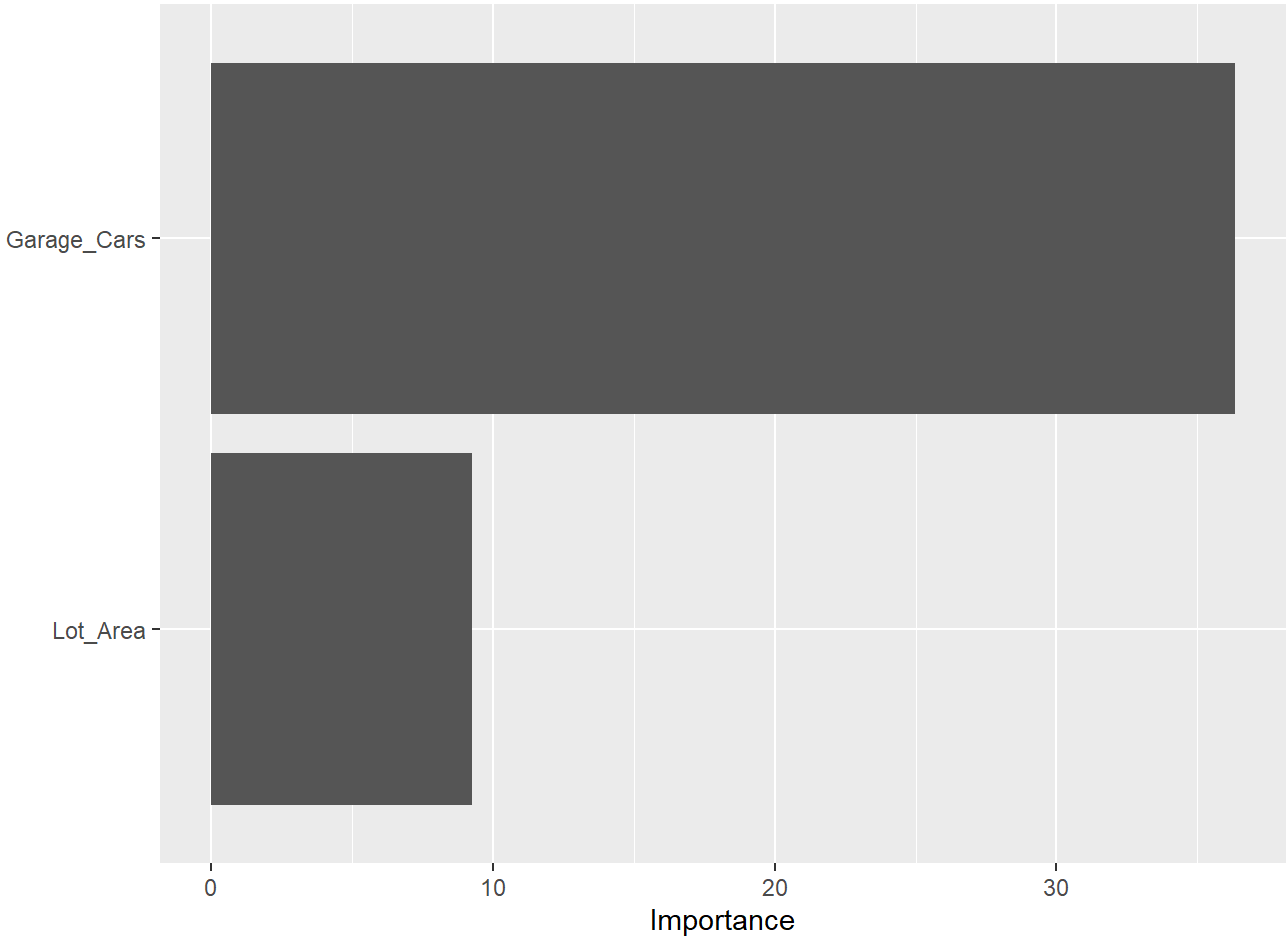
Variable importance assesses which predictor variables that have the strongest predictive value to the outcome. In this case, we can make the following observations:

- Gross Living Area was the number one predictor with neighborhood, Ground living area with overall quality, and Garage car with lot area.


Gr_Liv_Area
MS_SubClassTwo_Story_PUD_1946_and_Newer
MS_SubClassOne_Story_PUD_1946_and_Newer
NeighborhoodStone_Brook
NeighborhoodNorthridge_Heights
NeighborhoodNorthridge
Bsmt_Unf_SF
MS_SubClassTwo_Story_1946_and_Newer
age
Bsmt_QualGood
Overall_QualVery_Excellent
NeighborhoodSomerset
Garage_Cars
NeighborhoodGreen_Hills
MS_SubClassDuplex_All_Styles_and_Ages
Overall_QualExcellent
Bsmt_QualTypical
NeighborhoodCrawford
Lot_Area
Overall_CondExcellent
0
5
10
15
20
Importance


Overall_QualVery_Excellent
NeighborhoodNorthridge
Overall_QualExcellent
NeighborhoodNorthridge_Heights
Overall_QualVery_Good
NeighborhoodStone_Brook
NeighborhoodCrawford
NeighborhoodClear_Creek
NeighborhoodOld_Town
NeighborhoodTimberland
Overall_QualGood
NeighborhoodBriardale
NeighborhoodIowa_DOT_and_Rail_Road
Overall_QualAbove_Average
NeighborhoodCollege_Creek
NeighborhoodGreen_Hills
NeighborhoodSomerset
NeighborhoodMeadow_Village
Overall_QualAverage
NeighborhoodVeenker
0
5
10
15
Importance


Gr_Liv_Area
Overall_QualVery_Excellent
Overall_QualExcellent
Overall_QualVery_Good
Overall_QualGood
Overall_QualAbove_Average
Overall_QualAverage
Overall_QualBelow_Average
Overall_QualFair
Overall_QualPoor
0
10
20
Importance


Garage_Cars
Lot_Area
0
10
20
30
Importance